Akanksha Sachan

Contact

Joint Carnegie Mellon University-University of Pittsburgh PhD Program — aks203@pitt.edu
Aging Institute & Dept. of Immunology, School of Medicine, UPitt — LinkedIn
Machine learning for Genomics & Epigenetics — Twitter | Github | Google Scholar

Education

University of Pittsburgh — *'24 – Present*
PhD Candidate in Computational Biology
Advised by Dr. Aditi Gurkar & Dr. Jishnu Das

Carnegie Mellon University — '22 – '24
PhD Student in Computational Biology
Comp-Bio Department, School of Computer Science | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay — '18 – '22
BTech. in Chemical Engineering | GPA: 8.06/10

Awards and Honors

- **American Federation for Aging Research** (AFAR) Graduate Scholarship — *'25*
- **Best Poster** Award at the Aging Institute Research Day, UPitt — *'24*
- **1st Place, Student A100 track** NeurIPS LLM Efficiency Challenge | team of 5 — *'23*
- **$500 Track Prize** (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 — *'22*
- Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay — *'22*
- Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship — *'21*
- Awarded a full scholarship ($5,000) for iSURE undergraduate research program by the University of Notre Dame — *'20*
- **Top 0.2%** (Rank 3188 out of **1.1M+** candidates) in JEE (Main) — *'18*
- **Top 1.5%** (Rank 2517 out of **160k+** candidates) in JEE (Advanced) — *'18*

Invited Talks

"FIREFate: Functional and Interpretable Regulatory Encodings of cell Fate" | *paper-in-prep*

- Keystone Symposia: Artificial Intelligence in Molecular Biology — *Sep '25*
- International Society for Computational Biology (ISCB)/(ISMB) — *Jul '25, '26*
- American Association of Immunologists (AAI) Annual Meeting 2026 — *Apr '26*
- Center of Lung Aging and Regeneration, School of Medicine, UPitt — *Nov '25*
- 30th Annual School of Medicine Graduate Student Research Symposium, UPitt — *Oct '25*
- Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat — *Aug '25*

"Metabolic rewiring of skeletal muscle driven by loss of DNA-repair" | *paper-in-prep*

- Center for Systems Immunology Annual Retreat, UPitt — *Nov '25*
- 23rd Annual Department of Immunology Scientific Retreat, UPitt — *Sep '25*
- Aging Institute Research Day, UPitt — *Nov '24*

"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"

- 50th European Calcified Tissue Society (ECTS) Congress — *Apr '23*

Athletic Accolades

Bronze Medal | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay *'19*
3rd Position | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU *'23*
Swam 12-hours | Swimathon organized by the IIT Bombay aquatics committee *'19*
Ran 26.2 miles | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*
Ran 13.1 miles | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*

Leadership & Societal Contributions

Vice President *Aug '25 – Present*
Allegheny Science Policy and Governance (ASPG)
Leading a recognized chapter of the **National Science Policy Network**; bridged scientists and PA legislators on policies about **AI/data-centers and Aging** at the State Capitol, Harrisburg.

Teaching Assistant *Fall '23 – Spring '24*
02-760: Lab Methods for Computational Biologists, SCS, CMU
Taught a lecture on **genomics technologies** and gene regulation to a class of ∼20 students.

Coordinator *Aug '23 – Aug '25*
Technights, Carnegie Mellon University, Robotics Institute
Led a **computer-science outreach** program teaching middle-school girls data analysis, foundation models, and deep learning.

Team Lead *Mar '24*
4D Nucleome Consortium Hackathon, Seattle, Washington
Led a team of 5 to build integrative pipelines for **single-cell HiC** analysis from cell-atlas-scale datasets; delivered a 30-minute presentation to consortium researchers.

Treasurer • Senator *Aug '23 – Aug '25*
Graduate Student Association (CPCB GSA)
Managed a **$5,121** budget across CMU and UPitt events; organized the 2024 open-house for admitted CPCB students.

Editorial Board Member • Design Convenor *'19 – '21*
Insight, the student newsletter of IIT Bombay
Sketched & illustrated **1 Flagship** & **1 Special Edition** magazine cover and a merch t-shirt; contributed editorial oversight and article curation.

Convenor, Girls Swim Team *'19 – '20*
Aquatics, Indian Institute of Technology Bombay
Led logistics, training coordination, and community-building events for the institute girls' team.

Consortiums & Workshops

SenNet Consortium Annual Meeting *Sep '25*
Cellular Senescence Network, NIH, Washington, DC
Attended talks on cellular senescence atlasing and organ-specific senescence signatures.

Grace Hopper Celebration *Oct '24*
AnitaB.org, Philadelphia
Talks and brainstorming sessions with tech-industry leaders.

Computational Genomics Summer Institute *Jul '23*

University of California, Los Angeles

Delivered the lightning talk "Connecting 3D genomic organization to gene expression."

Grad Cohort for Women *April '23*

Computing Research Association (CRA-WP), San Francisco

Mentoring sessions with women in CS across industry and academia.

Research Experience Prior to Thesis

Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture *'22 – '24*

Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU

Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous cell lines; implemented **hypergraph-based representation learning** for integrative analysis of 3D genome architecture in single-cell, multi-omic mouse-brain datasets.

Translational Mechanobiology Lab, MBI, National University of Singapore *'22*

Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine

Built a **Generative-Adversarial-Network** (GAN) pipeline for colorectal-cancer imaging to support robust clinical deep-learning deployment; reviewed domain-adaptation and distribution-shift methods for liver-fibrosis staging from annotated medical images.

Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich *'21*

Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech | Presentation

Added a **Type-2 Diabetes** pathway to an agent-based **osteoporosis** model and ran large-scale simulations on the **CSCS Swiss National Supercomputing Center**; validated against in-vivo data from 235 diabetic patients and identified osteoid production as a key parameter via **sensitivity analysis**.

Multicellular Systems Engineering Lab, University of Notre Dame *'20*

Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering | Poster

Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs (OpenCV, Fiji) to study organism-wide signaling; evaluated variance under gene-knockout in the Ca^{2+} signaling pathway using PCA features.

Key Coursework

Machine Learning & Statistics: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

Predicting genomic-hub mediated TF-perturbation response via geometric deep learning — '23

10-708: Probabilistic Graphical Models | Code Repo

Augmented the GEARS GNN for Perturb-seq response prediction with a Hi-C contact graph (K562 **cancer cell-line** Micro-C, 20 kb bins mapped to gene TSSs), adding 3D-genome topology to the co-expression and Gene Ontology graph features.

Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs — '23

02-710: Computational Genomics | Code Repo

Developed a **GNN** to predict methylation from HiC data, connecting structure to phenotype; clustered **single-nucleus DNA methylation** data with K-means and t-SNE.

Efficient Online Continual Semi-Supervised Learning — '22

10-701: Intro to ML

Improved semi-supervised OCL-PDS with active learning and optimized pseudo-labeling; outperformed the ER-FIFO method on the Amazon-WPDS dataset.